Omkar Padave

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Summary

Results-driven Machine Learning and AI Engineer with 4+ years of experience developing and deploying end-to-end AI/ML solutions across machine learning, deep learning, Generative AI, NLP, and computer vision. Experienced in building LLM-powered RAG applications, production ML pipelines, and scalable AI services. Strong background in model training, fine-tuning, semantic retrieval, MLOps, and deploying reliable AI solutions from development through production.

TECHNICAL

Languages: Python,Java,C++,SQL
AI/ML: PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, NLP, Computer Vision, Model Training & Evaluation, Fine-Tuning
Generative AI: LLMs, RAG, LangChain, LangGraph, OpenAI API, AWS Bedrock, AI Agents, Prompt Engineering
Retrieval & Databases: Embeddings, Semantic Search, ChromaDB, Pinecone, PostgreSQL, MySQL, MongoDB
Backend & MLOps: FastAPI, REST APIs, MLflow, LangSmith, Model Monitoring, CI/CD
Cloud & DevOps: AWS (SageMaker, EC2, S3, Lambda), Azure, Docker, Kubernetes, Terraform, Git

EXPERIENCE

ML Developer – AI Systems Jun 2022 – Aug 2024
Konsultera Solution Pvtd Ltd

• Engineered end-to-end AI/ML solutions using Python, PyTorch, TensorFlow, and Scikit-Learn for NLP, recommendation, and predictive modeling use cases; optimized model training and evaluation workflows to improve prediction accuracy and production performance.
• Built Generative AI and RAG applications using LLaMA 2, OpenAI, Google Gemini, LangChain, ChromaDB, and Pinecone, enabling semantic retrieval and context-aware responses across structured and unstructured enterprise data.
• Established production ML workflows with Docker, CI/CD, MLflow, and model monitoring, streamlining model versioning, deployment, experiment tracking, and operational reliability across AI applications.
• Developed scalable model-serving services using FastAPI and REST APIs, integrating AI models with MySQL and MongoDB and deploying inference workloads across AWS Bedrock, EC2, Lambda, Azure, and Hugging Face Spaces.
Created computer vision and document intelligence pipelines using OpenCV, ViT, Swin Transformer, Donut, PaddleOCR, and spaCy to automate document classification, OCR, information extraction, and named entity recognition.

ML & Data Engineering Apr 2020 – May2022
Konsultera Solution Pvtd Ltd

• Built and optimized deep learning models using CNNs, RNNs, and Transformers for user behavior analysis, image understanding, and recommendation systems.
• Developed real-time computer vision pipelines using YOLOv8 and U^2-Net for object detection and image segmentation in production workflows.
• Applied transfer learning, fine-tuning, and hyperparameter optimization to improve model accuracy, inference performance, and scalability.
• Integrated ML pipelines with cloud-based APIs and data workflows to enable secure, low-latency predictions and campaign analytics.

PROJECTS

ATS Resume LLM App
Designed and deployed an LLM-powered ATS resume optimization application using Python, OpenAI APIs, and Google Gemini Pro, implementing prompt engineering and automated resume-to-job-description matching workflows. Deployed the solution on AWS to provide scalable resume analysis and optimization, improving ATS match scores by 50% and increasing users' chances of securing interviews.

Advanced Q&A Chatbots with DataStax Databases and Vector Embedding
Engineered and deployed an internal company Q&A chatbot by integrating DataStax Astra DB with vector embeddings and semantic search to provide employees with accurate, context-aware answers from enterprise knowledge sources. Implemented retrieval-based workflows and deployed the application using Hugging Face Spaces, reducing manual information lookup and human intervention by 40% while improving response accuracy and internal productivity.

Transformer-Based Document Classification & Extraction
Built a multimodal document intelligence system using Vision Transformers (ViT), Swin Transformer, Donut, PaddleOCR, and spaCy NER for document classification, OCR, and entity extraction, achieving **90%+ accuracy and precision** across structured and unstructured datasets.

EDUCATION

University of Dayton, Dayton, OH *Aug 2024 – May2026*
Master's in Computer Science

University of Mumbai, Mumbai, India *Sep 2019 – Jul 2022*
Bachelor's in Computer Science